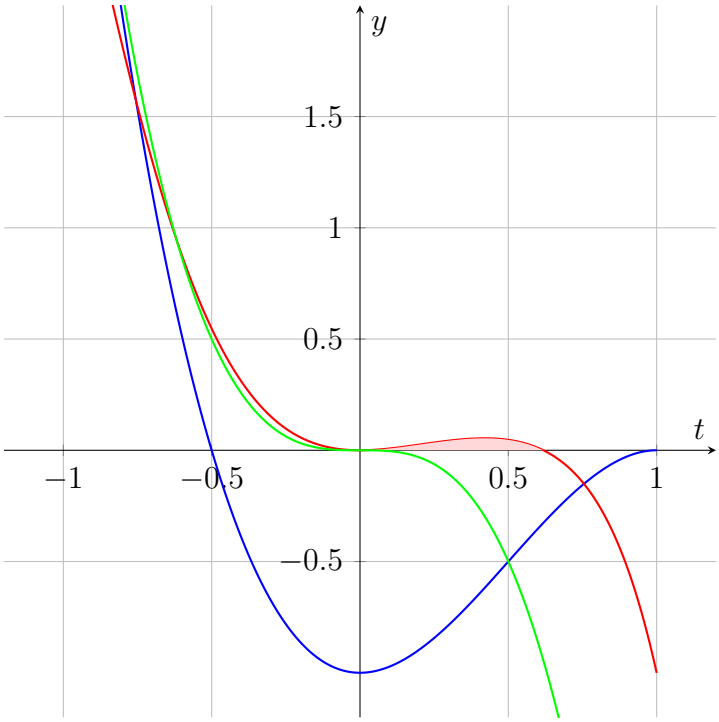
Ritangle 2026

Stage 1 - Question 2

Project π^*

October 21, 2026

Topic - Inequalities



*Solved by Nihal Sharma.

1 Problem

The problem statement was as follows:

> *Four points, A, B, C and D, are constrained to move on the x-axis.*
> *As t varies from $t = -1$ to $t = 1$, the points are in positions given by*
>
> 1. $A = t^2 - 1$
> 2. $B = t^3$
> 3. $C = t(t - 1)$
> 4. $D = t(t + 1)$
>
> *The diagram shows the case when $t = 0.4$*

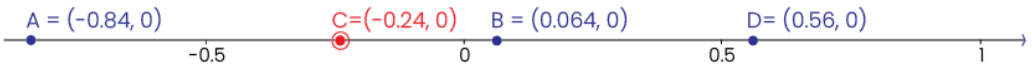


Figure 1: Reference image

For what proportion of the interval $-1 < t < 1$ is exactly one of the points closer to $(0, 0)$ than C is. Give your answer to 3d.p

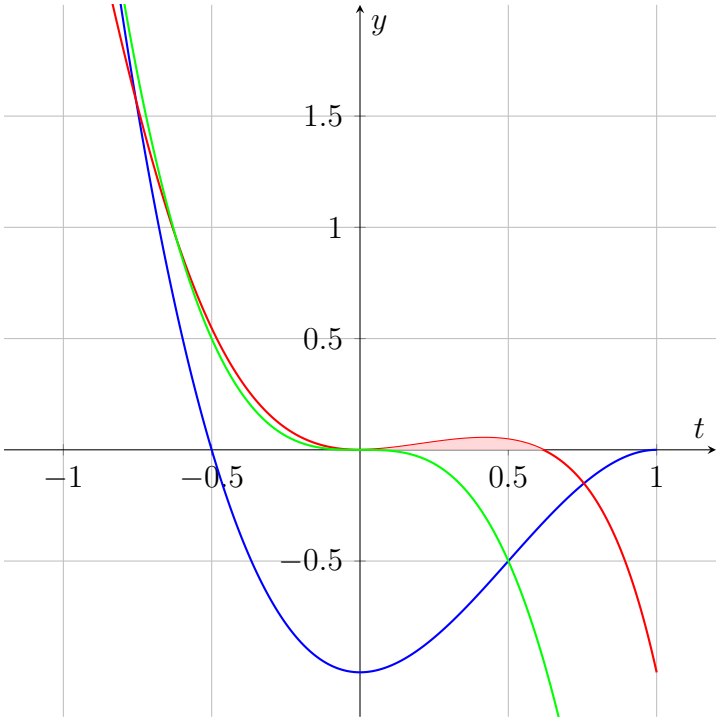
2 Solution

The key insight for this question is to recognise that using the distance from the origin is the same as using the square of the distance which avoids negative cases Let a, b, c, d be the squares of the distances of A, B, C, D from the origin:

$$a = |A|^2, \qquad b = |B|^2, \qquad c = |C|^2, \qquad d = |D|^2.$$

For an interval to satisfy our condition, exactly one of the following must hold:

$$c - a > 0,$$
$$c - b > 0,$$
$$c - d > 0.$$



Because exactly one of the inequalities must be true, we are looking for the region for which only one curve is above the axis. From inspection we can tell this is the region from $0 < t < \alpha$, where α is some real constant

Hence we are looking for the smallest positive solution of $t^2(t-1)^2 - t^6 = 0 \implies t^2(t-1-t^2)(t-1+t^2) = 0 \implies t = 0, \frac{-1\pm\sqrt{5}}{2}$. However we require t to be strictly positive, hence $t = \frac{-1+\sqrt{5}}{2} \approx= 0.61803$

Because the interval $-1 < t < 1$ has length 2, the proportion of the interval that satisfies the conditions is approximately 0.309

Remark: This could very quickly be solved using graphical calculators such as desmos with domain restrictions

Final Answer Summary:

Proportion of interval = 0.309 $\implies$ *Hint: B=14*